Eastern Oyster (*Crassostrea virginica*)

Summary Scores

- **Sensitivity:** Very High
- **Exposure:** Low
- **Vulnerability:** Moderate

Sensitivity

Sensitivity Attributes	Score	LH DQ	Tally
Adult Mobility	4	3	
Complexity in Reproductive Strategy	3.8	2.56	
Dispersal of Early Life Stages	2	3	
Egg and Larval Survival and Settlement Requirements	3.52	2.7	
Historical Variability in Temperature	2.64	2.8	
Juvenile Habitat Requirements	2.88	3	
Other Stressors	4	2.9	
Population Growth Rate	1.96	1.92	
Prey Specificity	1.48	3	
Relationship to Calcified Organisms (due to impacts from Ocean Acidification)	3.88	2.9	
Spawning Cycle	2.44	3	
Stock Size/Status	3	1.5	
Total Sensitivity | Certainty	**Very High (4) | 1**		

Exposure

Exposure Factors	Mean	SD	Tally
Bottom Temperature	1.9	0.99	
Bottom Oxygen	1.69	0.81	
Bottom Salinity	2.35	0.89	
Bottom Aragonite Solubility	1.29	0.50	
Net primary production	1.11	0.23	
Downward Flux of Particulate Organic Carbon	1.26	0.46	
Total Exposure	**Low (1.31)**	**0.66**	

Vulnerability

Vulnerability	Mean	SD	Tally
Total Vulnerability	**Moderate (5.25)**	**2.63**	

Low Moderate High Very High

Figure 1: Summary Sensitivity, Exposure, and Vulnerability Scores and Score Distributions for Eastern Oyster. Exposure and Vulnerability scores are averaged spatially across the species distribution and within stocks if applicable (see Figure 5 and Figure 4). Life History Data Quality (LH DQ) and Standard Deviations (SD) are reported for senstivity and exposure/vulnerability scores, respectively.

Changes from Previous Assessment(s)

Table 1: Changes to Vulnerability, Exposure, and Sensitivity scores relative to Previous Assessments. If a citation for the previous assessment is not provided, it comes from Hare et al. (2016).

Metric	Previous Assessment(s)	Current Assessment	Change
Sensitivity	High	Very High	Increase
Exposure	Very High	Low	Decrease
Vulnerability	Very High	Moderate	Decrease
Directional Effect	Negative	Neutral	Increase

Stocks

Eastern Oyster is managed as a single stock or stock information was not considered for this iteration of the CVA. ## Score Descriptions:

Sensitivity:

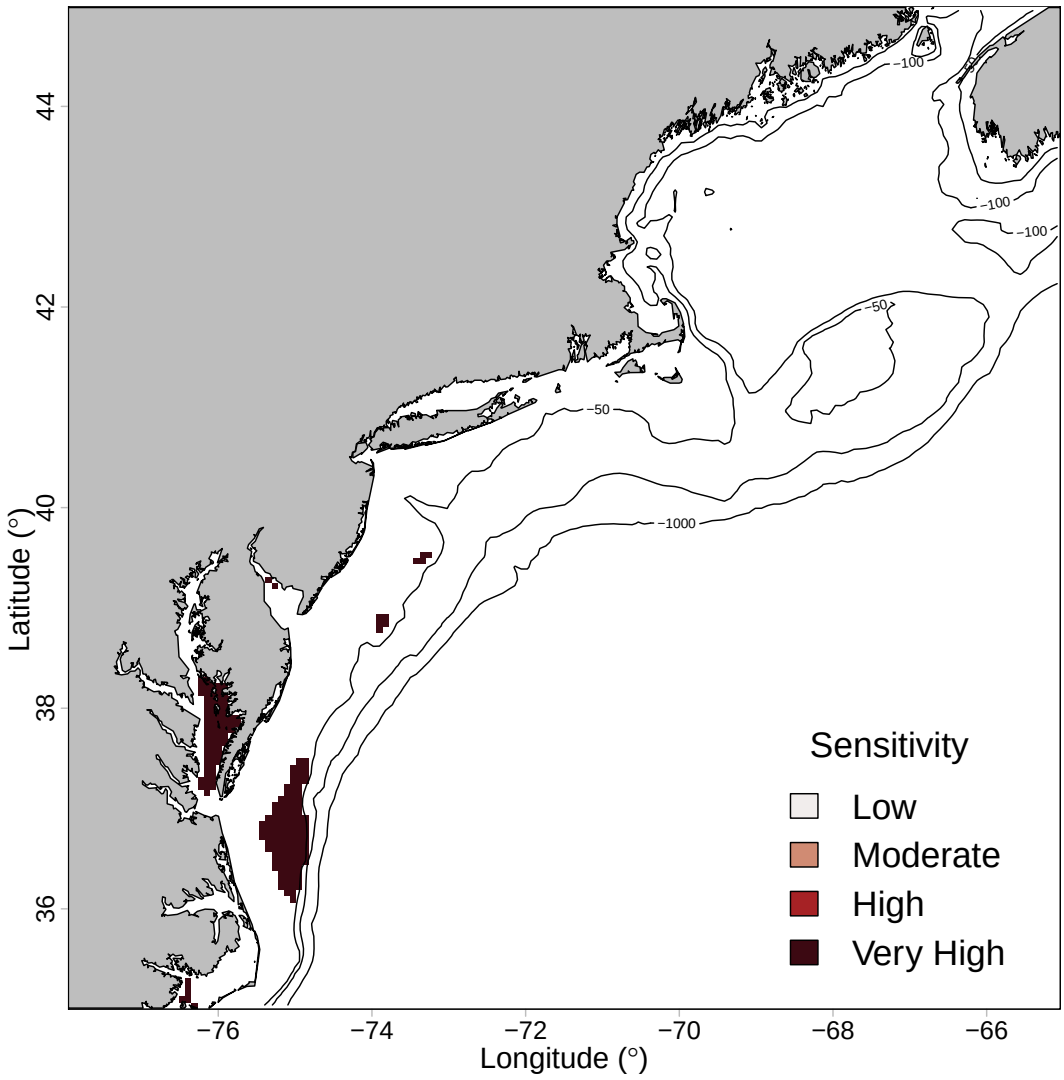


Figure 2: Eastern Oyster Sensitivity on the Northeast Shelf

The Eastern oyster received a **Very High** sensitivity score, with 100% certainty according to bootstrap analysis (Figure 1). Five attributes had scores greater than 3.5: Adult Mobility (4), Other Stressors (4), Relationship to Calcified Organisms (3.9), Complexity in Reproductive Strategy (3.8), and Egg and Larval Survival and Settlement Requirements (3.5) (Figure 1).

Eastern Oysters are sessile once they settle out of the plankton, and settle in estuarine habitats. As a result, they are exposed to a variety of stressors due to urbanization impacts such as hypoxia, loss of seagrass beds, dredging, and changes to freshwater inputs (Guyant et al. 2022). Spawning is initiated by warmer temperatures. Settling out of the plankton also is dependent on temperature and the presence of conspecifics. Because recruitment is density dependent, population collapse can be sudden.

In the 2016 Assessment, Eastern Oysters received a High sensitivity score. This was driven by Sensitivity to Ocean Acidification, Adult Mobility, and Other Stressors, similar to our results (Hare et al. 2016).

Exposure:

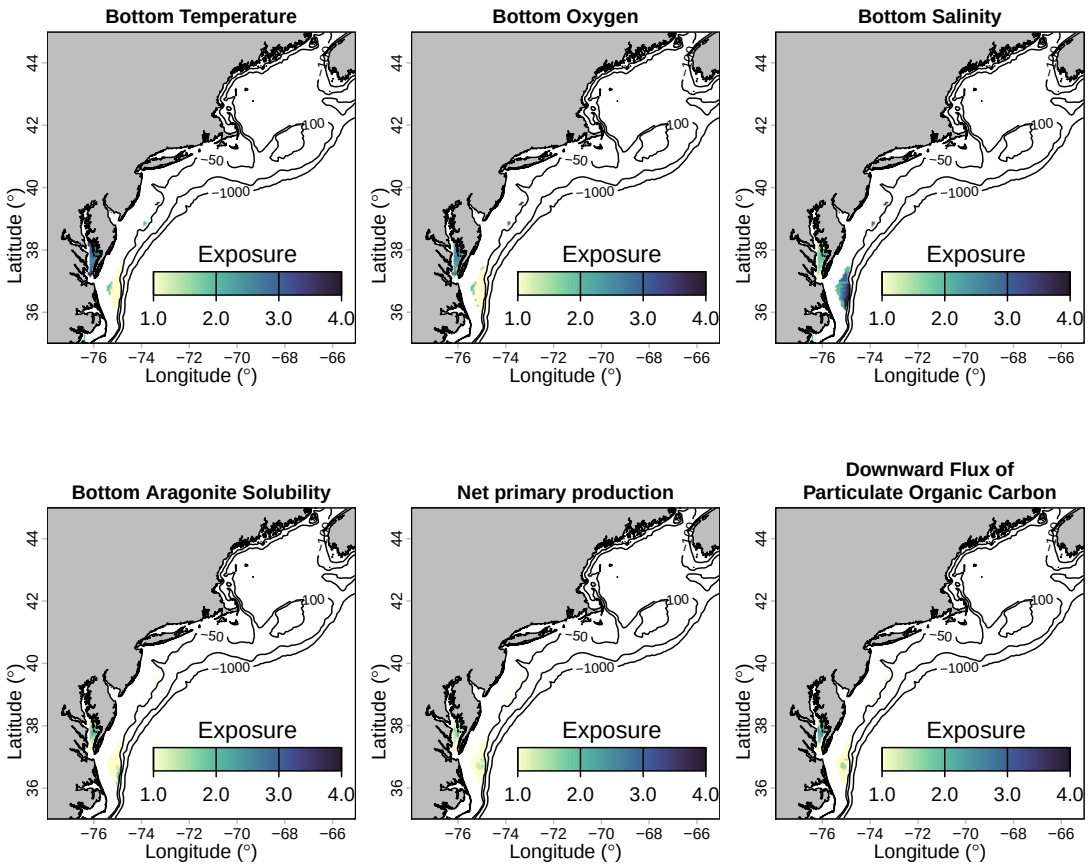


Figure 3: Maps of each Exposure Factor for Eastern Oyster. Exposure was calculated between 2014-2023 and 2025-2035

Mean exposure for Eastern Oyster was **Low** (1.33 +/- 0.66) (Figure 1). Scores were Moderate within Chesapeake Bay but were Low on the continental shelf (Figure 4). This was driven by high exposure to bottom temperature and oxygen changes within the Bay (Figure 3).

In the previous assessment, eastern oysters received a Very High exposure score, driven by sea surface temperature, air temperature, and ocean acidification (Hare et al. 2016).

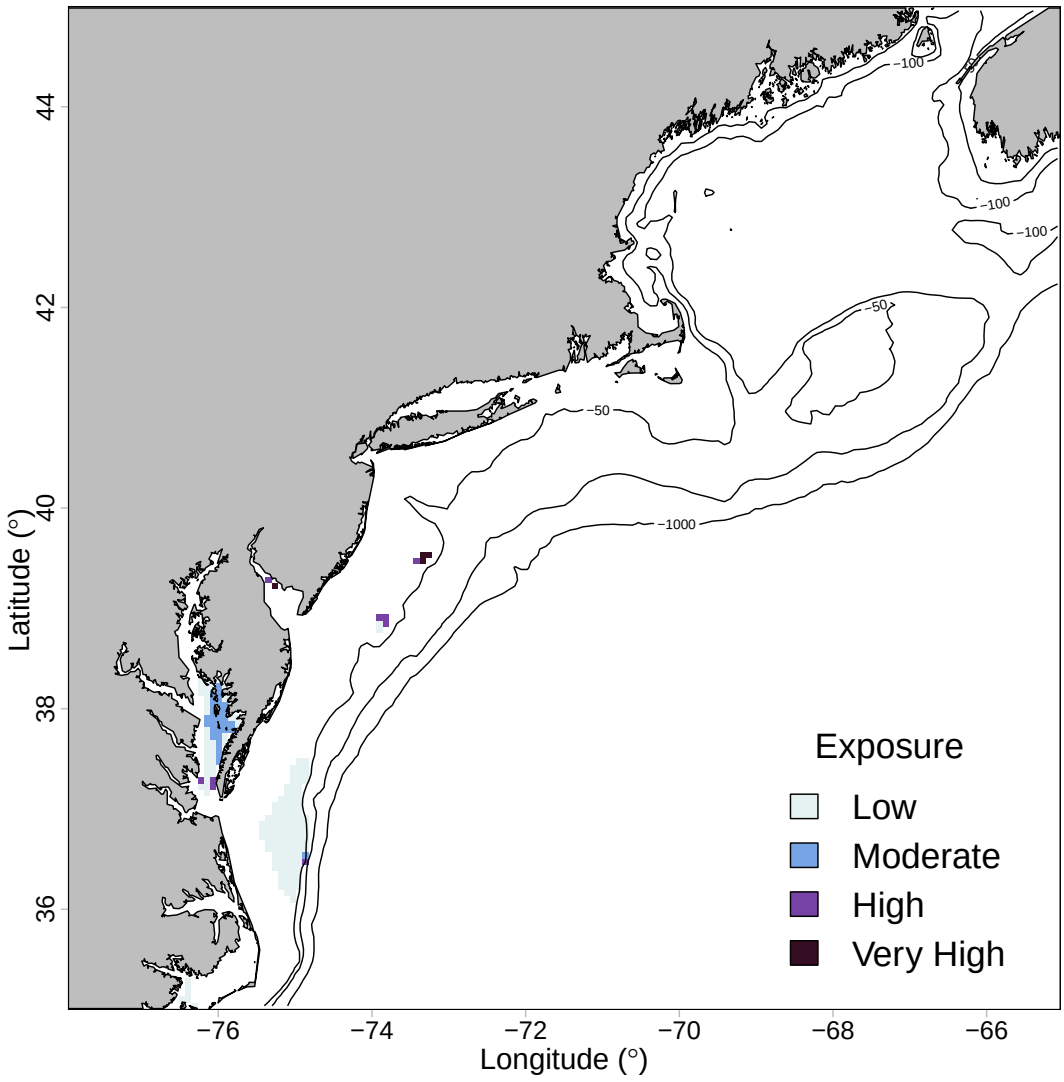


Figure 4: Total Exposure on the Northeast Shelf for Eastern Oyster. Total Exposure is calculated by applying the logic rule from Hare et al. (2016) within grid cells across exposure factors (see Figure 3)

Vulnerability:

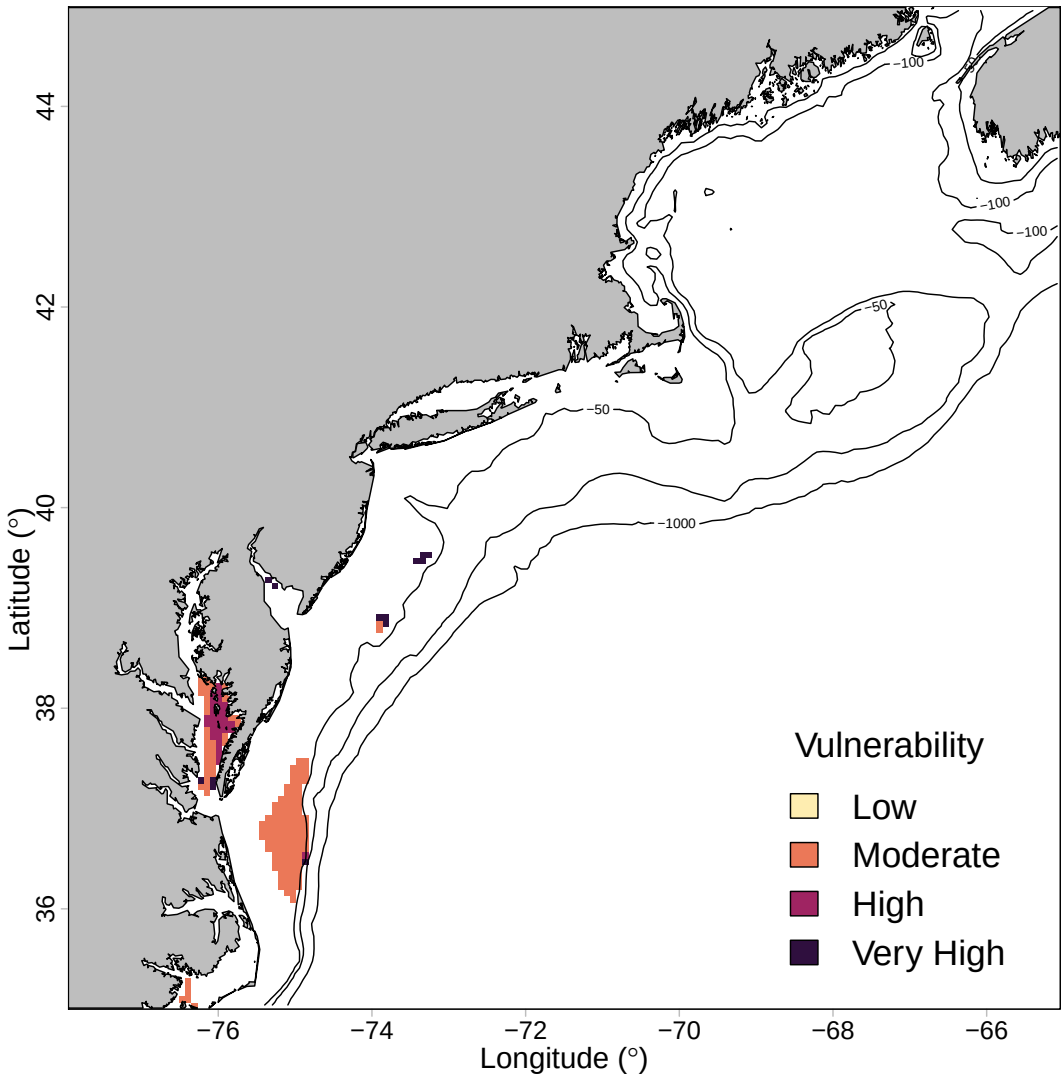


Figure 5: Vulnerability on the Northeast Shelf for Eastern Oyster. Vulnerability is the product of sensitivity and exposure (see Figure 4 and Figure 2)

Mean vulnerability for Eastern Oyster was **Moderate** (5.25 +/- 2.63) (Figure 1). Vulnerability was highest in the Chesapeake Bay (Figure 5).

In the previous Assessment, Eastern Oyster received a Very High Vulnerability score, driven by high and very high sensitivity and exposure scores, respectively (Hare et al. 2016).

Data Quality Metrics

Life History Data Quality:

90% of data quality scores were greater than or equal to 2, indicating excellent data quality. Population Growth Rate and Stock Size/Status were identified as data gaps (mean attribute data quality score < 2)

(Figure 1).

Model Confidence Score:

The eastern oyster ensemble model received a 1.2 (+/- 0.9) mean model confidence score, indicating low confidence in the model. Poor sample size drove down model confidence scores and the model struggled to predict nearshore habitats. The ensemble AUC score for the eastern oyster model was 0.95, indicating that the model had excellent ability to predict oyster presence/absence. ## Additional Metrics

Distribution Shifts:

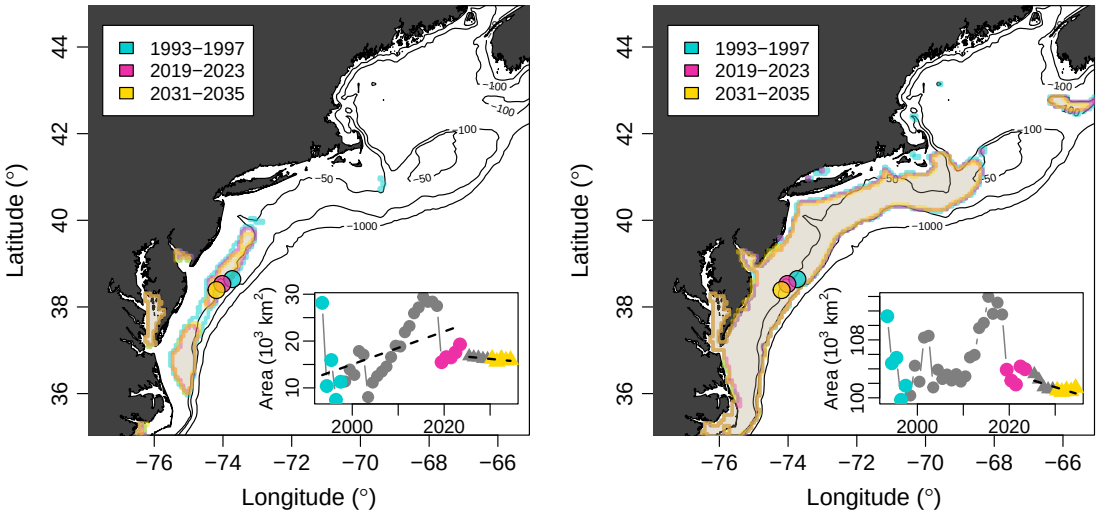


Figure 6: Distribution shifts on the Northeast Shelf for Eastern Oyster. Polygons illustrate core habitat (left) and species range (right) for five-year average periods. Points illustrate weighted centroid, calculated as the average position with species distribution models as weights. Inset timeseries illustrate the shift in core and habitat range area over time. Circles represent points from the MOM6 hindcast; triangles represent points from the MOM6 forecast. Dashed lines represent significant linear regressions

Note: For an animated version of this plot, please see the online version of this document.

Eastern oyster habitat was increasing between 1993 and 2015, but recently declined significantly (Figure 6). The weighted centroid shifted 28 km west-southwest.

Under near future climate conditions, these trends are predicted to continue. Further habitat loss is predicted across the species range and the weighted centroid is predicted to shift a further 21 km southwest (Figure 6).

In the previous Assessment, the eastern oyster received a Moderate Distributional Vulnerability Rank score (Hare et al. 2016).

Table 2: Distribution shifts based on five-year average distributions. Changes are calculated relative to the previous time period.

Time Period	Weighted Centroid Shift (km & bearing)	Core Habitat Area (change) (km^2)	Habitat Total Area (change) (km^2)	Latitude Range (change) (deg)	Change in Longitude Range (deg)
1993-1997	NA	24,144	109,499	5.24	7.06
2019-2023	28.3 km WSW	17,191 (-6,954)	103,316 (-6,182)	4.99 (-0.25)	6.90 (-0.16)
2031-2035	21.2 km SW	15,968 (-1,223)	101,048 (-2,268)	4.99 (0)	6.73 (-0.16)

Directional Effect:

Experts predicted that climate change would have a **neutral** effect on Eastern Oysters. Certainty was low at 60%. While oysters live in vulnerable estuarine habitats (Guyant et al. 2022), spawning occurs throughout the year when temperatures are warm (Berringan et al. 1991). Aquaculture programs also help contribute to population recovery (MacKenzie 1981). In the previous Assessment, experts predicted that climate change would have a negative impact on Eastern Oysters (Hare et al. 2016).

Climate Effects on Abundance and Distribution

Climate change, warmer temperatures, and perhaps different tidal ranges, may increase competition for resources with current competitors, like hooked mussel, invasive species such as *Didemnum* sp., or increase exposure to predators like veined rapa whelk (White and Wilson 1996). Depletion of oyster beds will also likely have negative effects for many marine species and habitats, for they provide refuge and homes for many fish and invertebrates species and help prevent erosion (Eastern Oyster Biological Review Team (EOBRT) 2007). Hurricanes can cover oysters in silt, wash them off beds, or impose extended "freshettes" on oyster beds to the degree that they are significantly impaired and need restoration. This may become more problematic, for hurricanes are thought to increase with increased warming temperatures and climate change (Perret et al. 1999). Since adult oysters are sessile and inhabit mainly highly vulnerable habitats in areas with high climate exposure, their spawning habitat is also highly vulnerable (Farr et al. 2021).

Life History Synopsis

Eastern oysters are found along the eastern U.S. from the Gulf of St. Lawrence to the Gulf of America. They inhabit brackish and salty waters, forming beds or reefs by attaching to each other or other hard substrate (Ozbay et al. 2014). Eastern oysters are a keystone species, and provide valuable habitat for benthic and intertidal organisms. Oysters are sessile filter feeders, contributing to increased water quality through reduced sediment suspension and nutrient uptake and sequestration (Ozbay et al. 2014). Eastern oyster populations in the Mid-Atlantic and Gulf declined in the 19th and 20th centuries, mainly from anthropogenic factors such as land use, habitat degradation, and overharvesting. Disease mortality from Dermo and MSX continues to diminish oyster recovery (Eastern Oyster Biological Review Team (EOBRT) 2007). Aquaculture production of oysters contributes to population recovery and habitat restoration programs across the East Coast (MacKenzie 1981). Oysters spawn during the summer in the North and Mid-Atlantic and throughout most of the year in the South Atlantic when temperatures remain above 20°C and salinities above 10ppt (Berringan et al. 1991). The first larval stage (trochophore) is formed

four to six hours following fertilization. Larvae normally settle two to three weeks after hatching, but can last up to 2 months (Eastern Oyster Biological Review Team (EOBRT) 2007). Larvae move both by passive and active (as development progresses) transport, although larval transport in estuaries seems to be minimal. A balance is reached where the larvae can find suitable areas to settle while still being retained in estuaries (Eastern Oyster Biological Review Team (EOBRT) 2007). Settlement ends at the end of the larval stage, and the larval duration is highly temperature dependent (Shumway 1996). Fritz et al. (1984) found that planktotrophic oyster larvae fed upon a variety of phytoplankton taxa (coccoids, centrate diatoms, pennate diatoms, dinoflagellates, and flagellates). Baldwin and Newell (1991) found that the larvae are actually omnivorous, preying on both autotrophic and heterotrophic plankton. Predation on oyster larvae is extremely high by filter-feeders in the water column and in the benthos. Along the Atlantic coast, drills have been responsible for mortality on 33% of spat in Long Island Sound. Oysters can compete with other mussels and barnacles for shell and substrate spaces, resulting in increased mortality when competitors are abundant (Kennedy 1996). Juveniles and adults feed primarily on phytoplankton and suspended detritus, relying on food-laden water being moved past them in order to extract food from the water column (Langdon and Newell 1996). Maturity occurs between 1 and 3 years of age (Florida State University Marine Lab n.d.). Several ages have been proposed, the most common being 20 years. Oysters in Maryland appear to survive at least 10 years based on plantings at restoration sites which are not thought to have received natural spat sets. It is likely that the maximum longevity is substantially longer than 10 years (Maryland Department of Natural Resources 2025). Wild and aquaculture harvest is managed by individual jurisdictions or states, and management guidelines vary (Ozbay et al. 2014).

Supplemental Scores

Exposure Timeseries

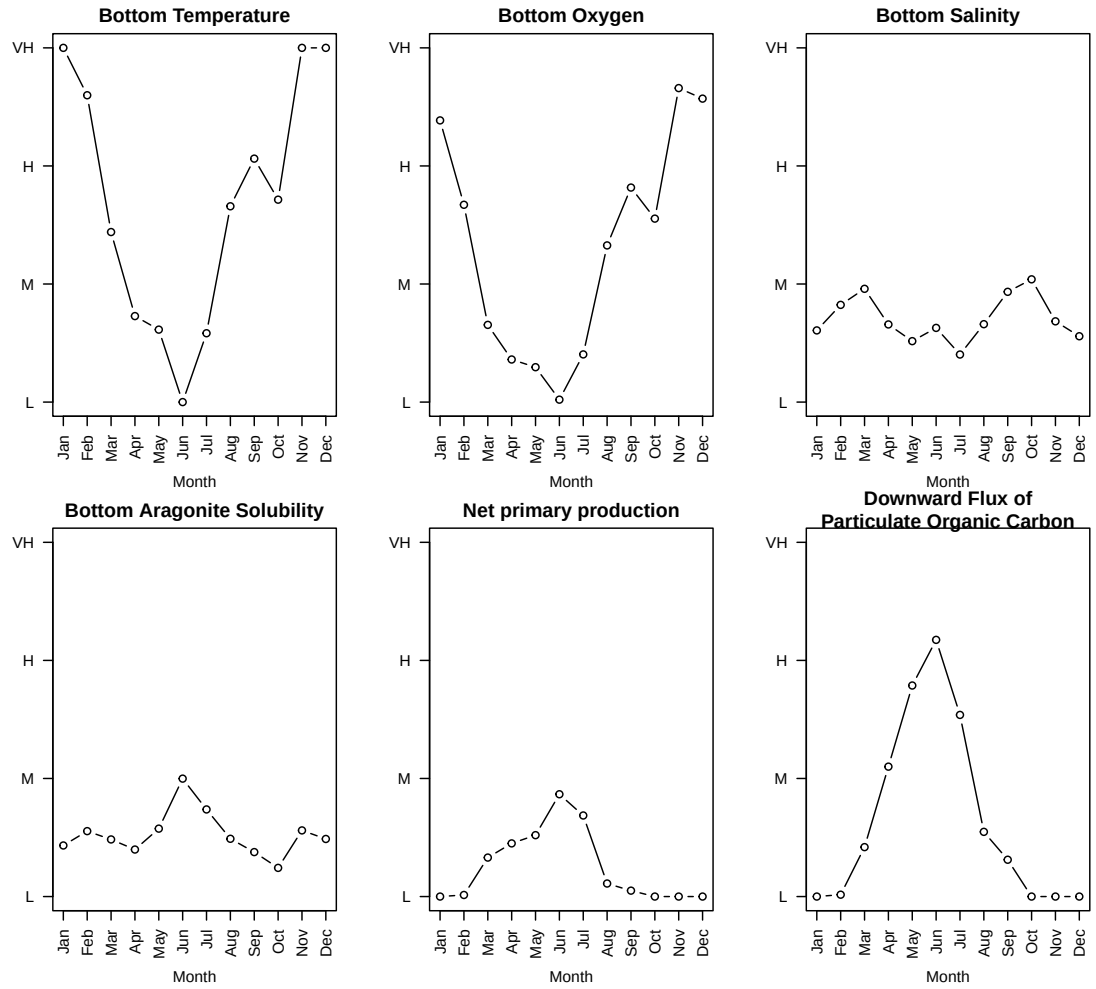


Figure 7: Monthly Timeseries of each Exposure Factor for Eastern Oyster. Exposure was calculated between 2014-2023 and 2025-2035

Exposure for the eastern oyster varied throughout the year (Figure 8). Exposure was Low from March through August due to low bottom temperature and oxygen exposure (Figure 7). For the remainder of the year, exposure was Moderate to High (Figure 8).

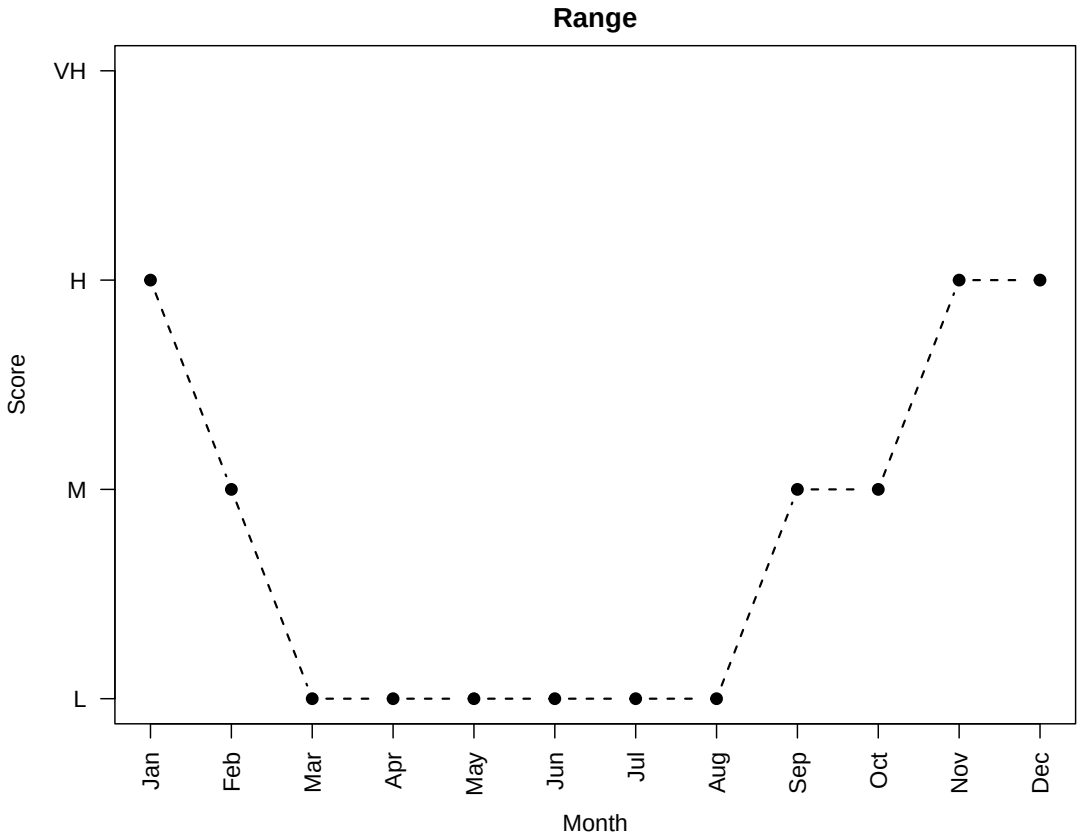


Figure 8: Timeseries of Total Exposure for Eastern Oyster

Exposure due to Significant Factors

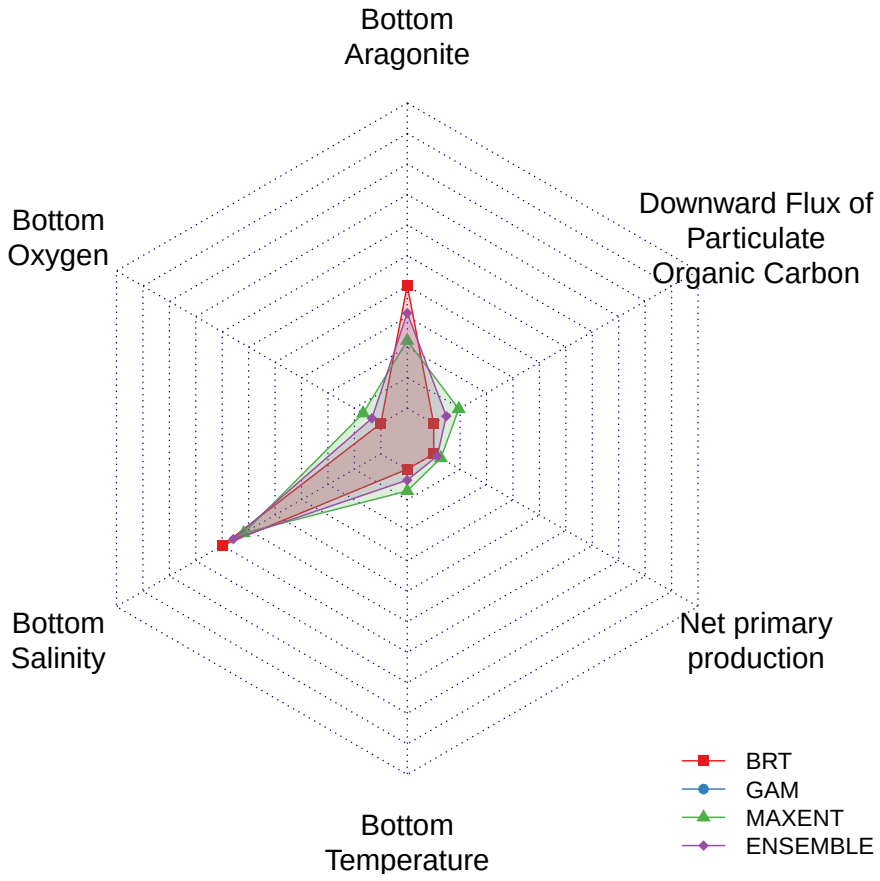


Figure 9: Dynamic Factors driving Eastern Oyster distributions. The closer the polygon is to the edge of the figure, the more important the variable is. Each ring represents 10% variance explained. Factors must explain at least 10% of variance in the ensemble model to be used to calculate total exposure.

Bottom salinity and aragonite solubility were important drivers of species distribution (Figure 9). Eastern Oysters live in estuarine habitats and build calcium carbonate shells (Ozbay et al. 2014). When these factors were used to calculate exposure, scores were Low (1.03 +/- 0.22) (Figure 11, Figure 10).

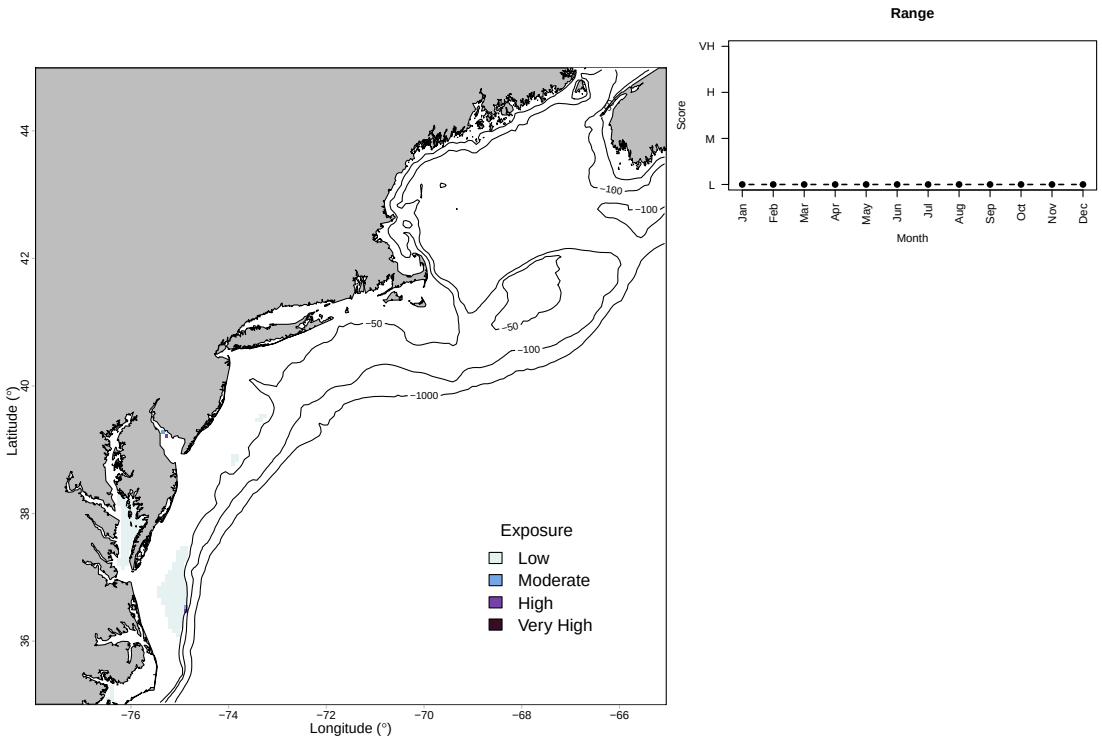
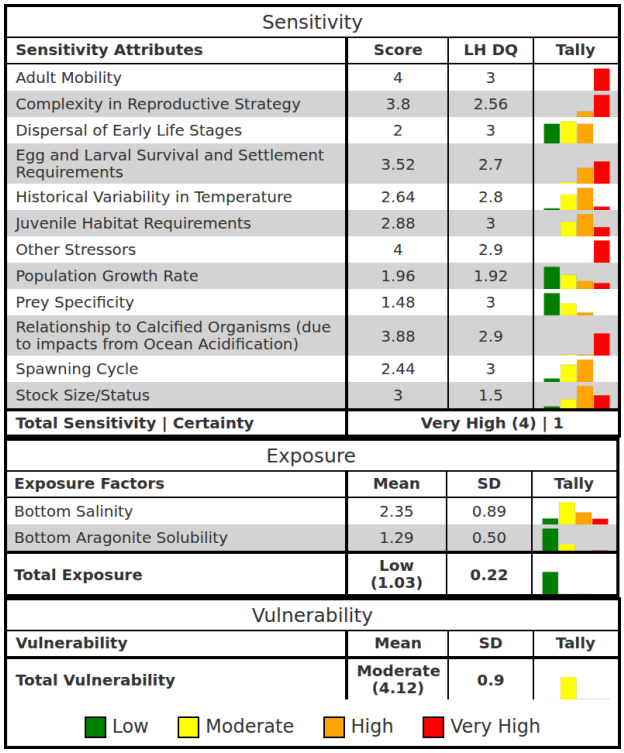


Figure 10: Total Exposure Map and Timeseries when Exposure was calculated with only Exposure Factors critical to Eastern Oyster distributions



Sensitivity

Sensitivity Attributes	Score	LH DQ	Tally
Adult Mobility	4	3	
Complexity in Reproductive Strategy	3.8	2.56	
Dispersal of Early Life Stages	2	3	
Egg and Larval Survival and Settlement Requirements	3.52	2.7	
Historical Variability in Temperature	2.64	2.8	
Juvenile Habitat Requirements	2.88	3	
Other Stressors	4	2.9	
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Prey Specificity	1.48	3	
Relationship to Calcified Organisms (due to impacts from Ocean Acidification)	3.88	2.9	
Spawning Cycle	2.44	3	
Stock Size/Status	3	1.5	
Total Sensitivity | Certainty	**Very High (4) | 1**		

Exposure

Exposure Factors	Mean	SD	Tally
Bottom Salinity	2.35	0.89	
Bottom Aragonite Solubility	1.29	0.50	
Total Exposure	**Low (1.03)**	**0.22**	

Vulnerability

Vulnerability	Mean	SD	Tally
Total Vulnerability	**Moderate (4.12)**	**0.9**	

Low Moderate High Very High

Figure 11: As in Figure 1 for Eastern Oyster, but with only significant Exposure factors and the resulting Vulnerability Scores.

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